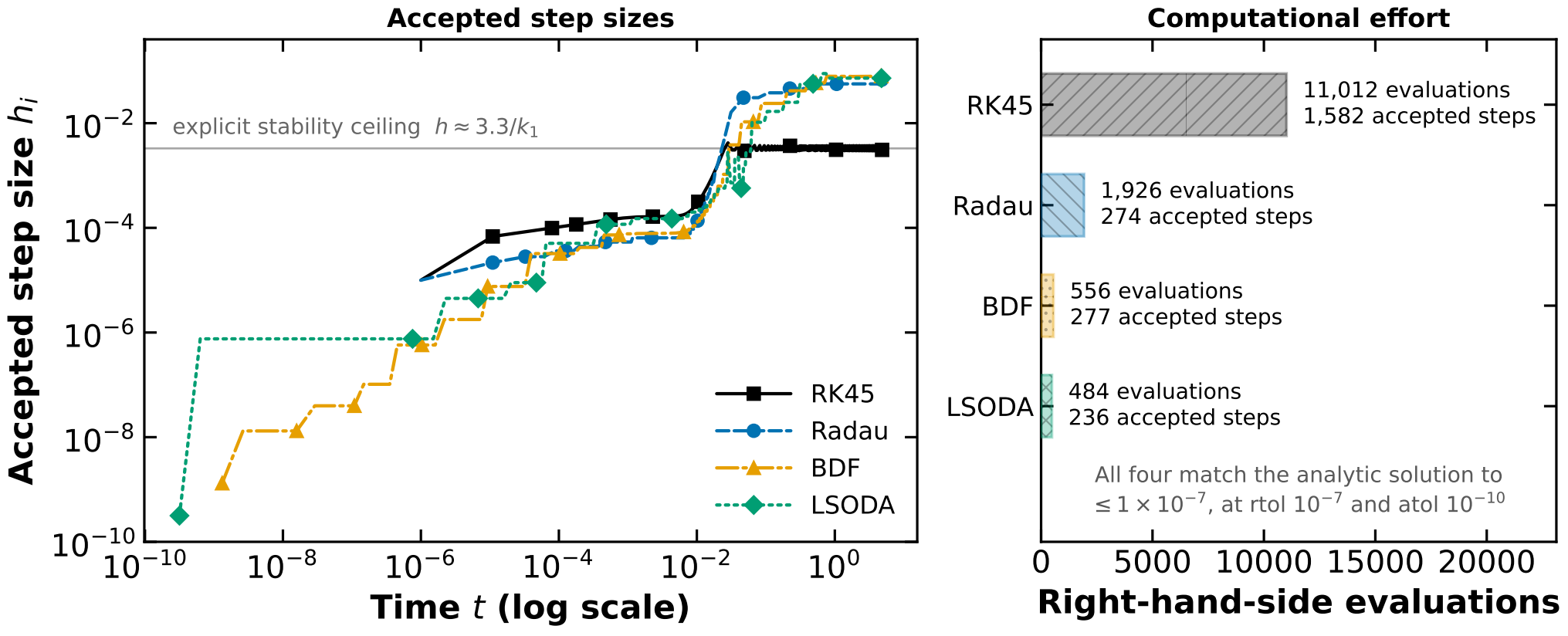

Accepted step sizes
explicit stability ceiling $h \approx 3.3/k_1$
RK45
Radau
BDF
LSODA
Accepted step size h_i
Time t (log scale)
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-10}
10^{-8}
10^{-6}
10^{-4}
10^{-2}
10^{0}
Computational effort
RK45
11,012 evaluations
1,582 accepted steps
Radau
1,926 evaluations
274 accepted steps
BDF
556 evaluations
277 accepted steps
LSODA
484 evaluations
236 accepted steps
All four match the analytic solution to
$\leq 1 \times 10^{-7}$, at rtol 10^{-7} and atol 10^{-10}
0
5000
10000
15000
20000
Right-hand-side evaluations